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REALISTIC CONTROL-VALVE PRESSURE DROPS

Here is a method of determining optimum control-valve pressure drop — sufficient pressure drop to assure automatic control, without wasting energy.

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Even in the most amicable of engineering departments, one of the most controversial subjects is the matter of how much pressure drop should be assigned to control valves. The confrontation is generally between the process designers, who realize that pressure drop consumes energy, and therefore should be minimized, and the instrumentation engineers, who know that it is the pressure drop that provides the driving force moving the fluid through the control valve, and that the greater the pressure drop, the greater the controller's ability to influence the process and change the value of the controlled variable.

In the extreme, if the pressure drop at the control valve were zero, the controller would have zero ability to change the value of the controlled variable and thus would have no effect. It is most curious that, despite the obvious importance of this aspect of automatic process control, so little has been published in the way of specific guidelines for determining the proper amount of control-valve pressure drop. "Proper" is that amount that will assure the successful functioning of the control system, yet avoid energy waste.

What the literature says

The control valve handbook of one leading valve manufacturer does not even discuss how to establish control-valve pressure drop. Another manufacturer's handbook suggests that the control valve often should have one-third of the total friction drop in the system. However, the handbook also points out that "under certain circumstances" 25% might be satisfactory, or possibly even 10% might be appropriate.

A statement that appears in Vol. 2 of the "Instrument Engineers' Handbook," revised ed., page 412, asserts that the amount of control-valve pressure drop that should be specifically assigned for control is none at all (0%). The implication is that process designers, in their usual charitable manner, can be depended upon to design "fat" into the system in the form of excess pressure drop. This "fat" will end up in the control valve and will provide all that is necessary for automatic control.

Arguing over arbitrarily chosen percentages of system pressure drop that should be given to a control valve is at best a waste of time, because there is no real engineering basis for the percentage numbers. What is needed is a method based on sound, credible principles. Here we introduce such a method for determining the optimum control-valve pressure drop, balancing the need to provide enough pressure drop to achieve control with the need to conserve energy.

Engineering the control-valve pressure drop

A typical situation is shown in Fig. 1. We will use this diagram not only to identify the problem but also to establish some basic truths and define some terms.

Assume that a process-design engineer is working on the charge circuit for a hydrocarbon feedstream that passes first through some preheat exchangers, then through a fired heater, and finally ends up in a fractionator. The

fractionator operates at 20 psig, and 15 psi is necessary to lift the liquid, from grade, up to the feed nozzle. At what is considered to be the design flowrate in this circuit (call this flowrate Q_d), there will be 32 psi of pressure drop in the preheat exchangers, 60 psi in the fired heater, and 30 psi in the piping. Since the intent is to have this stream on flow control, there will also be a 2-psi drop in the orifice meter.

Let us call the pressure drop taken up by the piping and all items of process equipment in the system, including the orifice, the "friction pressure drop," and designate it by the letter F. In this example, $F = 124$ psi. Having reached this point in the development of the design, the design engineer now approaches the instrument engineer with the question — "How much pressure drop is required for the control valve?" Once this figure is established, the design engineer can complete the pressure balance, determine how much pressure is needed at the start of the circuit, and draw up the specifications for a suitable pump.

To answer the question, the instrument engineer falls back on the popular rule that one-third of the system drop should be in the control valve. This would put the control-valve pressure drop at one-half of F, or 62 psi. The design engineer takes this number, calculates that the pump discharge pressure must be 221 psig, and says:

"No way. This would call for a big pump, with a bigger driver, that would consume far too much energy. Why does the pressure drop at the valve have to be so high?"

This usually puts the instrument engineer on the defensive. The response may be, "How much pressure drop can be allowed?" The process designer's answer: "How about 10 psi?" The instrument engineer knows intuitively that the control system cannot work over any significant range if the control valve has only 10 psi, but he or she lacks any soundly based argument to prove the point. Eventually, the two settle for 25 psi, and the instrument engineer buys the coffee and hopes for the best. Surely, there has to be a better way.

The friction pressure drop, F, has been defined as the pressure drop taken, at design flowrate Q_d, by the process equipment and the piping. Bear in mind that F must change if Q_d changes. Notice also that this system, like most, starts and ends at two points of relatively constant pressure. Let us call the beginning and end pressures for the system P_s and P_e. In Fig. 1, P_e is the fractionator pressure plus the static head, that is, 35 psig. P_s will be the pump discharge pressure, which would be 184 psig if the control valve were to be assigned a pressure drop of 25 psi. The phrase "relatively constant pressure" means that the pressures P_s and P_e do not change to any appreciable extent when the flowrate in the system changes. The difference between P_s and P_e provides the driving force for moving the fluid through the whole system. The pressure balance equation is:

$$P_s = P_e + F + \Delta P \qquad (1)$$

where ΔP is the control-valve pressure drop. Rearranging this yields:

$$\Delta P = (P_s - P_e) - F \qquad (2)$$

We should note carefully what Eq. (2) is telling us. One of the most aggravating questions that process designers can put to instrument engineers is, "Why can't you put in a larger control valve and take less pressure drop?" This is an out-and-out trap, and instrument engineers owe it to themselves to know how to avoid being drawn into it. The answer is obvious from Eq. (2). The pressure drop at the control valve can only be what is left of the overall system pressure drop after the friction pressure drop has been deducted.

Suppose that in the Fig. 1 example, the control valve size turned out to be 2 in., with a ΔP of 25 psi assigned. Let us then take out the 2-in. control valve and put a 3-in. one in its place. Would the pressure drop at the control valve change? Certainly not. The control-valve pressure drop must be what is left of $(P_s - P_e)$ after F is deducted — so it is still 25 psi. The 3-in. control valve would be pinched down in its stroke farther than the 2-in., but the pressure drop at a flowrate of Q_d would still be 25 psi. Put in a 4-in. control valve, and ΔP would still be 25 psi at the design flowrate Q_d, although the 4-in. valve would be pinched down even farther.

The bottom line is that the control-valve pressure drop has nothing to do with valve size. It is determined only by the pressure balance, Eq. (2). Period!

Fig. 1 reveals still one more important concept. Suppose

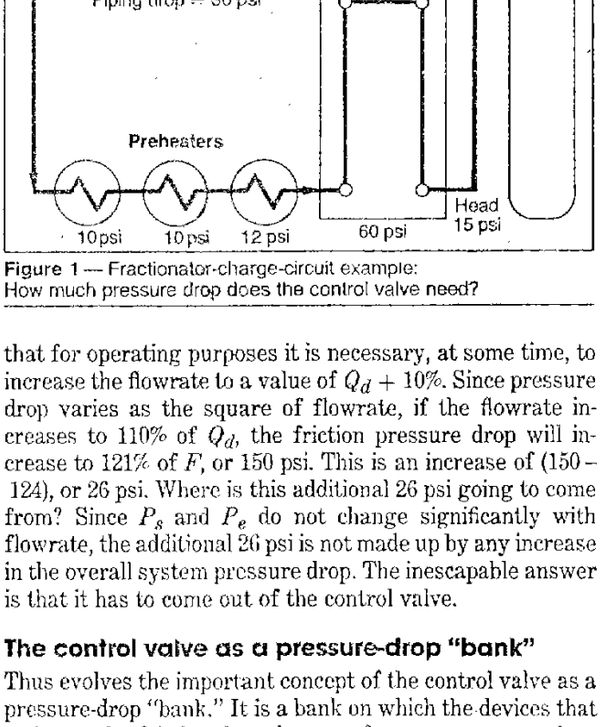


Figure 1 — Fractionator-charge-circuit example: How much pressure drop does the control valve need?

that for operating purposes it is necessary, at some time, to increase the flowrate to a value of $Q_d + 10\%$. Since pressure drop varies as the square of flowrate, if the flowrate increases to 110% of Q_d, the friction pressure drop will increase to 121% of F, or 150 psi. This is an increase of (150 – 124), or 26 psi. Where is this additional 26 psi going to come from? Since P_s and P_e do not change significantly with flowrate, the additional 26 psi is not made up by any increase in the overall system pressure drop. The inescapable answer is that it has to come out of the control valve.

The control valve as a pressure-drop "bank"

Thus evolves the important concept of the control valve as a pressure-drop "bank." It is a bank on which the devices that make up the friction drop draw on for more pressure drop when it is necessary to increase the flowrate in the system. It

also points out the direction that the engineering of the control-valve pressure drop should take. Note the word is "engineering".

The underlying principle is that, at the design stage, the control valve must be provided with enough pressure drop so that when the system flowrate is increased to the anticipated maximum rate, the control valve can give up the additional pressure drop demanded by the friction drop, and still have the minimum pressure drop left that will permit the automatic controller to keep the controlled variable on control.

In the Fig. 1 example, we saw that an increase in flowrate to 110% of Q_d would increase the friction drop F by 26 psi. This would have to come out of the control valve. If the control valve were assigned only 25 psi in the first place, then the flow control loop would be out of control at this higher flowrate. The control valve would go wide open, and the flowrate would be determined by the vagaries of the system.

In certain cases, there will be no need to perform calculations to determine control-valve pressure drop. One example is a steam-pressure-letdown station. The valve inlet and outlet pressures are the higher and lower steam pressures,

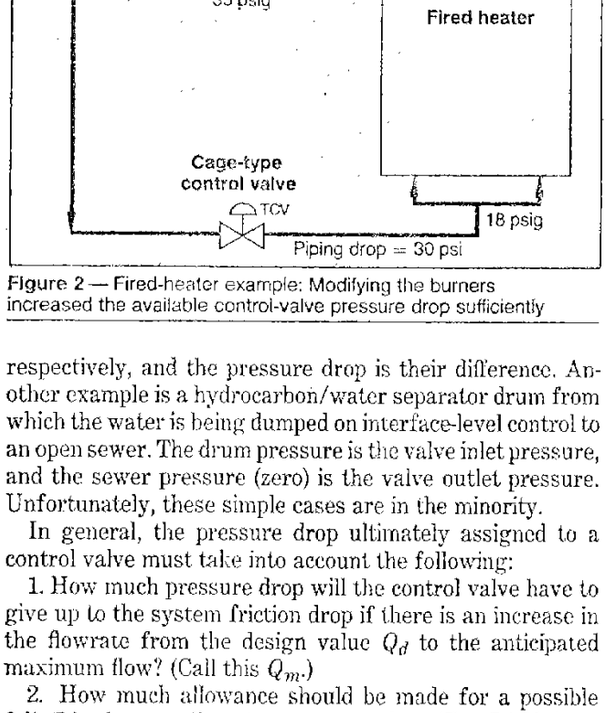


Figure 2 — Fired-heater example: Modifying the burners increased the available control-valve pressure drop sufficiently

respectively, and the pressure drop is their difference. Another example is a hydrocarbon/water separator drum from which the water is being dumped on interface-level control to an open sewer. The drum pressure is the valve inlet pressure, and the sewer pressure (zero) is the valve outlet pressure. Unfortunately, these simple cases are in the minority.

In general, the pressure drop ultimately assigned to a control valve must take into account the following:

1. How much pressure drop will the control valve have to give up to the system friction drop if there is an increase in the flowrate from the design value Q_d to the anticipated maximum flow? (Call this Q_m.)
2. How much allowance should be made for a possible falloff in the overall system pressure drop $(P_s - P_e)$ if there should be an increase in system flowrate?
3. Even in the wide-open position, a control valve will impose some resistance to flow; therefore it will create some pressure drop. This will be a function of the type of control valve used (plug, cage, ball, butterfly, etc.).

What we now need are methods for putting numbers on each of the three contributing factors just listed.

Allowance for increase in friction drop. To calculate this, we need to know the ratio of the maximum anticipated flowrate, Q_m, to the design flowrate, Q_d, or Q_m/Q_d. Note that the actual values need not be specified, only the ratio. Usually the process designer will be willing to estimate Q_m/Q_d, based on knowledge of how the plant will be run, or on the upper limit values of factors such as pump discharge rates. If the process designer is unwilling to commit to any estimate of the Q_m/Q_d ratio, then the instrument engineer will have to make a best estimate, based on what transients in flowrate may be expected as the control loop recovers, in its typical cyclic fashion, from an upset. At the end of this article, some suggested values for Q_m/Q_d are given.

At the design flowrate, Q_d, the friction drop will be F psi. At the maximum flowrate, Q_m, the friction drop will become:

$$F \times (Q_m/Q_d)^2$$

Nomenclature

B	Base pressure drop (control-valve pressure drop with valve in wide-open position)
F	Friction pressure drop at design flowrate
P_e	End pressure for the system
P_s	Beginning pressure for the system
ΔP	Pressure drop in the control valve
Q_d	Design flowrate
Q_m	Maximum anticipated flowrate

The increase in pressure drop, consequently, will be:

$$F \times (Q_m/Q_d)^2 - F, \text{ or } [(Q_m/Q_d)^2 - 1] \times F$$

The value of F, as determined by the process design engineers, is really a best estimate only. At this time, the plant will exist only on paper, and it will not be possible to get an accurate fix on pipe lengths, numbers and types of fittings, velocity, and physical properties of the stream. Therefore, it is probably prudent to apply a small factor of safety, 10%, to the above calculation. As a result, the allowance for the increase in friction drop will be:

$$1.1 \times [(Q_m/Q_d)^2 - 1] \times F \qquad (3)$$

Allowance for possible falloff in overall system pressure drop. A reasonable figure, considering that P_s and P_e are both supposed to be points of relatively constant pressure in the system, would be 10% of $(P_s - P_e)$. However, if this figure is used, it will prove practical in some cases but not in others.

Consider the situation in which fluid is flowing through a control valve from one vessel that operates at 300 psig, to a second vessel that operates at 250 psig. The overall system pressure drop would be 50 psi. If the allowance for loss of system driving force were 10% of this value (that is, 5 psi) it becomes obvious that the pressure in either of the vessels would only have to change 2%, for the allowance to be wiped out. Thus, this allowance should take into account not only

the system pressure drop ($P_s - P_e$) but also the pressure level at which the system operates.

The solution is to use the system end-pressure, P_s, as an indicator of the pressure level in the overall system, and divide the 10% allowance equally (5% and 5%) between the system pressure drop ($P_s - P_e$) and the system pressure level (P_e). The allowance (Item 2) therefore becomes:

$$0.05 \times (P_s - P_e) + 0.05 \times P_e$$

This expression simplifies to: $0.05\ P_s$ (4)

Allowance for the control valve itself. Even in its wide-open position, the control valve will impose some resistance to flow, and will require some base pressure drop. The value will vary with the design of control valve used, and probably with the application. This part of the control-valve pressure drop allowance will be identified by the letter B, for base pressure drop. By working with average line velocities, and assuming that in most cases the control valve will be one pipe size smaller than the line, and by averaging the calculated values of B over a range of valve sizes, the following values of B have been arrived at, and are recommended:

Control-valve type	B, psi
Single plug	11
Double plug	7
Cage valve (unbalanced)	4
Cage valve (balanced)	4
Butterfly	0.2
V-ball	1

Taking into account the pressure drop allowances that should be designed into the control valve for all three of the foregoing, the aggregate required control-valve pressure drop is therefore:

$$\text{Required } \Delta P = 0.05\ P_s + 1.1\ [(Q_m/Q_d)^2 - 1]\ F + B \quad (5)$$

A fired-heater example

A look at some examples may help to clear up any skepticism. Fig. 2 shows a temperature control valve in the fuel gas line to a fired heater. The system clearly starts at the fuel gas header, where the pressure is 35 psig; but where does the system end? At first glance, it appears to end at the inlet to the gas burners. However, the requirement for both the start and end points of the system is that they be points at which the pressure remains relatively constant despite changes in flowrate. This would not apply to the burner inlet because the pressure here varies considerably with changes in gas firing rate. The real end of this system is in the firebox, where the pressure is 0 psig.

The friction drop F at Q_d will be in the orifice plate (1 psi), the piping (2 psi), and the burner (18 psi). Thus, $F = 21$ psi. The available pressure drop is calculated from Eq. (2):

$$\text{Available } \Delta P = (P_s - P_e) - F\ P = 35 - 0 - 21 = 14 \text{ psi.}$$

The pressure drop required to accommodate the Q_m/Q_d ratio of 1.4 that has been specified is calculated from Eq. (5). The value of B for a cage control-valve is 4 psi. Therefore:

$$\text{Required } \Delta P = 0.05 \times 35 + 1.1 \times (1.4^2 - 1) \times 21 + 4 = 28 \text{ psi.}$$

These two calculations show that 28 psi is needed for the control valve, but only 14 psi is available. Furthermore, the required value, 28 psi, was not determined from some crystal ball percentage of the overall system drop. It was based on the shifts in pressure drop that the laws of nature call for.

Thus, the instrument engineer at last has an answer to the question, "So what if we can't give you 28 psi?" The answer is, "Unless the control valve has 28 psi pressure drop at the design flowrate Q_d, it will not be possible to increase the flowrate to anywhere near $1.4 \times Q_d$ ($= Q_m$) before the control valve is completely robbed of all of its pressure drop and the temperature-control loop goes out of control."

This example comes from an actual oil-refinery case. Engineering the required control-valve pressure drop, using Eq. (5), not only puts the comparison of required versus available pressure-drop on a sound basis but also provides a valuable side-benefit. If the requirement for pressure drop turns out to be much greater than what is actually available, as in this instance, an examination of the numbers that went into the Eq. (5) calculation will sometimes show which item is creating the need for so much ΔP. In this particular case, the big factor is the 18-psi pressure drop at the gas burners.

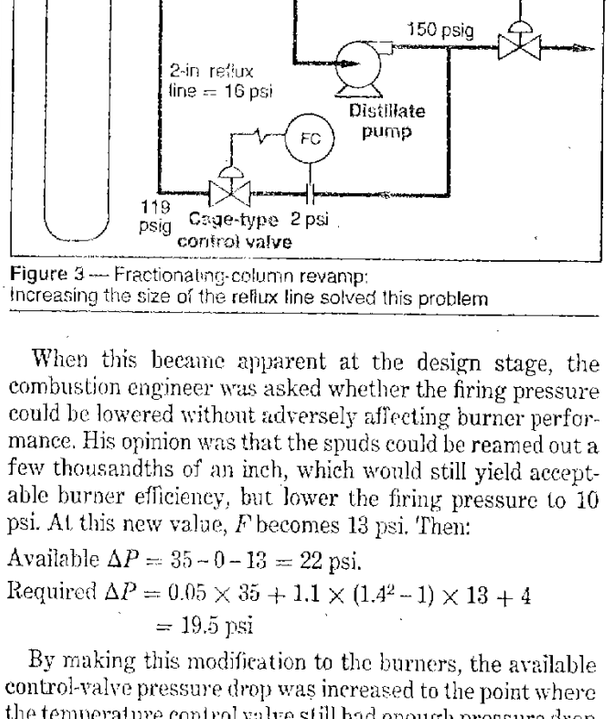


Figure 3 — Fractionating-column revamp: Increasing the size of the reflux line solved this problem

When this became apparent at the design stage, the combustion engineer was asked whether the firing pressure could be lowered without adversely affecting burner performance. His opinion was that the spuds could be reamed out a few thousandths of an inch, which would still yield acceptable burner efficiency, but lower the firing pressure to 10 psi. At this new value, F becomes 13 psi. Then:

$$\text{Available } \Delta P = 35 - 0 - 13 = 22 \text{ psi.}$$

$$\text{Required } \Delta P = 0.05 \times 35 + 1.1 \times (1.4^2 - 1) \times 13 + 4 = 19.5 \text{ psi}$$

By making this modification to the burners, the available control-valve pressure drop was increased to the point where the temperature control valve still had enough pressure drop left to keep the temperature loop on control even if the gas flowrate increased to 140% of the design rate.

A fractionating-column example

Example No. 2 (Fig. 3) concerns the revamp of a fractionating column. The process designers were not keen on purchasing a new distillate pump, although they knew higher reflux rates were needed. The question to the instrument engineer was, "Will the control valve have enough pressure drop?"

The system in which the reflux control valve was located started at the pump discharge, where $P_s = 150$ psig. The end of the system was in the fractionation column that was to operate at 100 psig, but 19 psi of static head was required to lift the reflux from grade up to the reflux nozzle. Hence, P_e was 119 psig. The friction drop F was 18 psi; 2 psi contributed by the orifice plate, and the rest by the piping.

The available pressure drop for the control valve, from the pressure-balance equation [Eq. (2)], was:

$$\Delta P = (150 - 119) - 18 = 13 \text{ psi}$$

The designers put the most likely value of Q_m/Q_d at 1.25.

Calculating the required pressure drop gave:

$$\text{Required } \Delta P = 0.05 \times 150 + 1.1 \times (1.25^2 - 1) \times 18 + 4$$
$$= 22.6 \text{ psi}$$

There was obviously not enough pressure drop to do the job. However, in examining the numbers used to calculate the required ΔP, it was clear that the biggest contributor was the piping drop, 16 psi. This seemed like an unusually large number, and the process designers were questioned about it. It turned out that they already had misgivings about the size of the reflux line, which was only 2 in. The instrument engineer's question simply confirmed their suspicions.

The decision was made to increase the reflux line size to 3 in. This lowered the piping pressure-drop to 4 psi, and the friction drop, F, to 6 psi. The second round of calculations produced an available pressure drop of 25 psi, and a required pressure drop of 15.2 psi. Once again, calculating the required control-valve pressure drop using this method not only showed that the situation as it stood was not feasible, but also suggested how to correct the problem.

Back to the original problem

For the final example, we go back to the situation in Fig. 1. This was for a new plant, and the haggling relating to the correct amount of pressure drop to be provided at the control valve would have been more appropriate for a primitive market place than for an engineering office.

Determining the right pressure drop in this case is more complex, but only slightly. The difficulty is that in order to use the required-pressure-drop equation, one has to know P_s. Unfortunately, P_s cannot be determined from the pressure balance until the control-valve pressure drop is known.

In this case, the system end-pressure, P_e, is set at 35 psig. The friction drop in the system is 124 psi (= F). Furthermore, the process designers are willing to agree on a value of Q_m = 120% of Q_d, so Q_m/Q_d is 1.2. The most suitable control-valve type would be be a cage valve, for which $B = 4$ [-].

From the pressure-balance equation [Eq. (2)]:

$$\text{Available } \Delta P = P_s - 35 - 124 = P_s - 159$$

From the required-pressure-drop equation [Eq. (5)]:

$$\text{Required } \Delta P = 0.05 \times P_s + 1.1 \times (1.2^2 - 1) \times 124 + 4$$
$$= 0.05\,P_s + 64$$

Equating these two expressions for the control valve ΔP produces a system start pressure, P_s, of 235 psig, and a control-valve pressure drop of 76 psi.

Notice that this number is even greater than the pressure drop that would be arrived at by assigning a third of the system pressure drop to the control valve. Yet, 76 psi is the pressure drop that must be in the control valve at the design flowrate for this system, if it is to be possible to increase the flowrate by the specified amount, 20%.

Remember that a pressure drop of 25 psi was the design engineer's best offer, and that it was accepted by the instrument engineer against better judgment. If 25 psi were actually selected, then P_s becomes 184 psig. Using these two numbers in the required-pressure-drop equation, the value of Q_m/Q_d can be back-calculated. It turns out to be 1.04! In other words, with a 25-psi pressure drop at design flowrate, the system flowrate could only be increased 4% before the control valve would go wide open for lack of pressure drop, and the control loop would go out of control.

Cleaning up the loose ends

This essentially completes the description of the method, but some loose ends should be dealt with.

First, process designers will usually calculate the value of the friction pressure drop, F, at the design flowrate, Q_d. Occasionally, however, they will calculate F at Q_m. This doesn't happen often, but it happens often enough that one should always check with the designers to verify the flowrate at which F was calculated. If it was calculated at Q_m, it should be scaled down by the factor $(Q_d/Q_m)^2$.

Second, if no commitment can be obtained from the process designer regarding the Q_m/Q_d ratio, then the suggested values for calculating the required control-valve pressure drop are 1.1 for flow-control valves, and 1.25 for level-, pressure-, and temperature-control valves. These numbers are based purely on anticipated flowrate transients as the control loop recovers from a disturbance.

Third, if the required-pressure-drop method described here is used, and the control-valve size turns out to be line size, this is no cause for concern. It may be an indication that the piping is on the small side, but the control loop will be able to do the job expected of it.

Last, Eq. (5) for calculating the required control-valve pressure drop will work equally well whether the numbers inserted are all in psi or all in kPa.

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